

MUCHINTHALA VISHNU VARDHAN NAIDU

📍 Bangalore, Karnataka

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Summary

B.Tech CSE graduate specializing in applied Machine Learning and Generative AI. Built and deployed a multi-agent RAG pipeline using LangChain, FastAPI, and FAISS, and shipped ML models via production REST APIs. Proficient in Python, PyTorch, FastAPI, Django, and SQL. Seeking an entry-level Machine Learning / GenAI Engineer role.

Technical Skills

- **Programming Languages:** Python, JavaScript, HTML5, CSS3
- **AI/ML & GenAI:** LLMs, RAG, LangChain, Prompt Engineering, Vector Databases (FAISS), PyTorch, Hugging Face Transformers
- **Libraries:** NumPy, Pandas, Scikit-learn, XGBoost
- **Frameworks & Tools:** FastAPI, Flask, Django, DRF, Streamlit, Git, GitHub, Postman, VS Code, Jupyter Notebook
- **Cloud & Databases:** Render, Docker, MySQL, MongoDB
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, OS, REST API Design, Machine Learning

Work Experience

ThopsTech Career Solutions

Bangalore, India

AI & Machine Learning Intern

Dec 2025 – Present

- Worked with machine learning models in Python using Scikit-learn, including data preprocessing, exploratory analysis, and model evaluation on real-world datasets.
- Applied supervised learning techniques such as regression and classification, and compared model performance using accuracy and other evaluation metrics.
- Used Pandas and NumPy to clean and prepare structured datasets for training and evaluating machine learning models.

Talent Trek E-Learning

Bangalore, India

Data Science Intern

Jun 2025 – Aug 2025

- Processed and analyzed datasets using Python, Pandas, and NumPy, performing end-to-end data collection, cleaning, exploratory analysis, and visualization.
- Created 5+ data visualizations and applied machine learning methods to extract insights that supported analytical decision making.

Projects

SolarNet Pro — Solar Radiation Prediction System 🐙 GitHub | *Python, Flask, Random Forest, XGBoost, REST API*

- Built a full-stack ML web app predicting solar radiation using Random Forest & XGBoost on real meteorological datasets with optimized hyperparameters.
- Deployed a production-grade Flask REST API with CORS handling, input validation, and end-to-end model serving pipeline.
- Implemented a multi-layer backend fallback system to ensure uninterrupted predictions during API failures.
- Designed an interactive frontend with Chart.js visualizations showing real-time predictions, model confidence, and feature importance.

Multi-Agent Research Assistant 🐙 GitHub | *Python, FastAPI, LangChain, FAISS*

- Designed a 4-agent RAG pipeline (Planner, Researcher, Summarizer, Report Generator) that decomposes research queries into sub-tasks and synthesizes a structured report, with per-agent error isolation so a single agent failure doesn't crash the request.
- Built a document ingestion pipeline (PDF/DOCX/TXT → chunking → FAISS vector store) using Gemini API embeddings to enable grounded question-answering over user-uploaded documents.
- Implemented automatic 4-provider LLM fallback (Gemini, OpenAI, Groq, DeepSeek) so the app stays responsive during rate limits or provider outages, instead of failing the request.
- Wrote a pytest suite covering pipeline orchestration and API validation; deployed the application live on Render.

Education

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| • NBKR Institute of Science and Technology
B.Tech in Computer Science and Engineering; CGPA: 8.19/10 | 2022 – 2026 |
| • Sri Chaitanya Junior College
Intermediate; Percentage: 86.8% | 2020 – 2022 |

Achievements

- **Certifications:** Full Stack Development (Simplilearn SkillUp), AI & Machine Learning (INNOVATE by HEXAIND), Data Science (Talent Trek E-Learning), Artificial Intelligence (AICTE).